

Auteur: Rob Willemsen
Studierichting: Informatica
Oefeningenreeks 2E nr. 18.6

$$\begin{aligned}4 \cdot \log_x(\sqrt{3}) + \log_x\left(3^{\frac{1}{3}}\right) + \log_x\left(\frac{1}{9}\right) + \frac{1}{3} &= 0 \\ \Rightarrow 4 \cdot \log_x\left(3^{\frac{1}{2}}\right) + \log_x\left(3^{\frac{1}{3}}\right) + \log_x(3^{-2}) + \frac{1}{3} &= 0 \\ \Rightarrow 2 \cdot \log_x(3) + \frac{1}{3} \cdot \log_x(3) - 2 \cdot \log_x(3) + \frac{1}{3} &= 0 \\ \Rightarrow \frac{1}{3} \cdot \log_x(3) + \frac{1}{3} &= 0 \\ \Rightarrow \frac{1}{3} \cdot \log_x(3) &= -\frac{1}{3} \\ \Rightarrow \log_x(3) &= -\frac{1}{3} \cdot \frac{3}{1} \\ \Rightarrow \log_x(3) &= -1 \\ \Rightarrow x^{-1} &= 3 \\ \Rightarrow x &= \frac{1}{3}\end{aligned}$$

References